B option command

1-7 Review of Mathematical Theory

Theory of Computation

Let us discuss each one by one -

)Negation

horizontal bar on letter.

S	1s
T	F
F	T

ii Conjunction (AND

If A and B are two propositions then conjunction of A and B is denoted by A^B. The truth table for this connective is

A	B	A^B
T	T	T
T	F	F
F	T	F
F	F	F

i) Disjunction (OR)

If A and B are two propositions then disjunction of A and B is denoted by AvB. The truth table for this connective is

A	B	AvB
T	T	T
	F	T
	T	T
F	F	

v) Implication

A and B are two propositions then if A then B is a proposition. It can be denoted byABorA=B.

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Theory of Computation

The truth table for this connective is		AB
	B	
A		
		T
	T	
		F
T	F	
	T	T
F		
		T
F	F	

1.
5.

For example: Find truth value for following proposition. refle

If an apple is not green then it must be red. Let A denotes An apple i not green" and B denotes It must be red The statement A is false and B is true. Hence the proposition is true.

v f and only if

f A and B are two propositions then A if and only if B. It can be denoted as AB The truth table will be Equival

A	B	AB
T	T	T
T	F	F
F	T	F
F	F	T

For example: I will go to the school if and only if it is not rainingHere statement A E

is 7 will go to the school The statement B is It is not rainingThus the resultant proposition is false.

1.5 Relations Solutior

Relaonship is a major aspect between wo objectseven this is true in our real life. Hen One ohect can be related with the other object by a some relation.Then those two objects forma pair based on this certain relationship. elements. DefinitionThe relation R is a collction for the set S which represents the pair of Som

For example:(a b is in R. We can represent their relation as a R b.The first By a

pair is chosen from a set called range. For

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Properties of Relations

A relation R on set S is 1.Reflexive if iRi for all i in S. 2.Irreflexive if iRi is false for all i in S. 3.Transitive if iRj and jRk imply Rk 4.Symmetric if iRj implics jRi 5 Asymmetric if iRj implies that jRi is false. Every asymmetric relation must be irreflexive. For exampleifA=ab|then reflexive relation R can be=a.ab, b| irreflexive relation R can be =a, b1 transitive relation R can be =(a.b)b, aa, a| symmetric relation R can be=a, b)b, a asymmetric relation R can be=a b

Equivalent Relation

A relation is said to be equivalence relation if it is reflexive, symmetric and transitive over some set S. Suppose R is a set of relations and S is the set of elements. For example S is the set of lines in a plane and R is the relation of lines intersecting to each other. A relation is said to be antisymmetric if a, b) e R and b, a e R then a =a.

Example 1.5: Determine whether R is equivalence relation or not where

A=0121R=001011)2, 2)2, 1

Solution The R is reflexive because 0, 01, 1)2, 2E R.

Where as R is not symmetric because (0, 1) is not in R whereas (1.0) is in R. Hence the R is not a equivalence relation.

Closures of Relations

Sometimes when the relation R is given, it may not be reflexive or transitive By adding some pairs we make the relation as reflexive or transitive. For exampleLetabbca, abb)